

1	(a) power = energy / time = (force × distance / time) = kg m ² s ⁻² / s = kg m ² s ⁻³	C1 C1 A1	[3]
(b)	(i) units of L^2 : m ² and units of ρ : kg m ⁻³ and units of v^3 : m ³ s ⁻³ ($C = P / L^2 \rho v^3$) hence units of C : kg m ² s ⁻³ m ⁻² kg ⁻¹ m ³ m ⁻³ s ³ or any correct statement of component units argument / discussion / cancelling leading to C having no units	C1 M1 A1	[3]
	(ii) power available from wind = $3.5 \times 10^5 \times 100 / 55$ (= 6.36×10^5) $v^3 = 3.5 \times 10^5 \times 100 / (55 \times 0.931 \times (25)^2 \times 1.3)$ $v = 9.4 \text{ m s}^{-1}$	C1 C1 A1	[3]
	(iii) not all kinetic energy of wind converted to kinetic energy of blades generator / conversion to electrical energy not 100% efficient / heat produced in generator / bearings etc (there must be cause of loss and where located)	B1 B1	[2]